Strategy-Space Modeling for Interpretable Quant Research

From Function-Space Model Selection to Live GLD Allocation

Joon Chun

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Starting Point

- The project started from a mathematical question, not from a trading rule.
- In class, we studied model selection using polynomial basis functions:

$$1,\ x,\ x^2,\ \ldots,\ x^d$$

- That raised the main idea:

Can trading strategies also be treated as basis functions?

- If a strategy maps a market state into a numerical score, then it can be evaluated across dates and placed into a design matrix.

Function Space Analogy

Concept	Polynomial Model	Strategy Model
basis function	$1, x, x^2, \ldots$	strategy signal functions
data point	x_i	market state at date t_i
matrix entry	$\phi_j(x_i)$	strategy score $s_j(t_i)$
model selection	choose degree / penalty	choose strategy basis / penalty
target	observed y_i	future cumulative return

Core design matrix:

$$A_{ij} = \phi_j(\text{market state at } t_i)$$

Here, each column is a strategy function evaluated over time.

Research Question

Main Question

Can a matrix of strategy signals explain or predict future GLD returns?

- First, define a strategy space numerically.
- Then test whether linear operators on that space have predictive value.
- Finally, translate predictions into portfolio exposure.

Important distinction:

$$\text{building a valid strategy space} \neq \text{proving every linear combination is better}$$

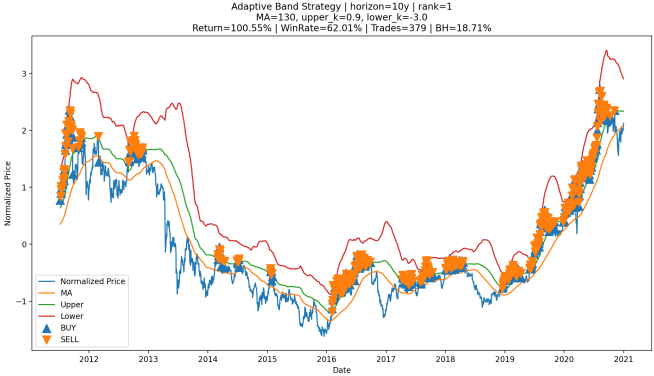
Four Initial Strategy Families

- **Adaptive Band**
  - price location relative to rolling normalized bands
- **Moving-Average Crossover**
  - trend state from short and long moving averages
- **Adaptive Volatility Band**
  - volatility state from high-low-close structure
- **Fear-Greed Candle Volume**
  - event-style signal from candle body, volume, and price zone

Strategy 1: Adaptive Band

Signal type: continuous

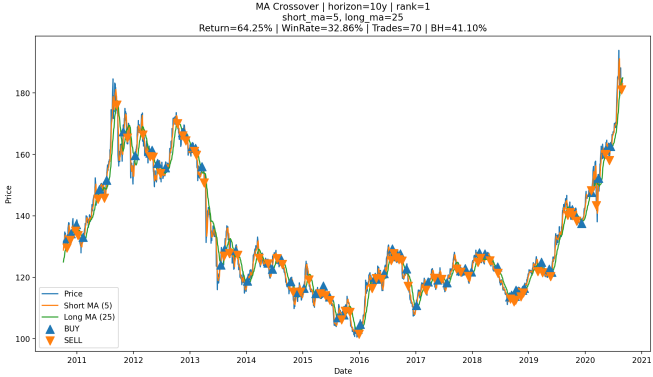
- Price is normalized relative to a rolling moving average
- Upper and lower bands are placed at scaled thresholds around the MA
- Score reflects how far price sits from the band midpoint
- Buy zone: price near or below lower band
- Sell zone: price near or above upper band



Strategy 2: Moving-Average Crossover

Signal type: continuous

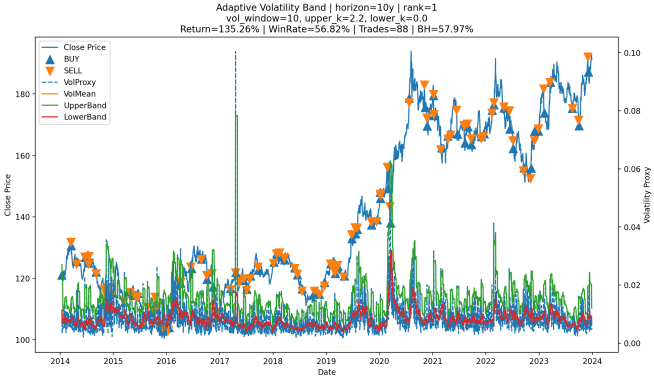
- Compares a short-window MA against a long-window MA
- Positive spread indicates uptrend; negative spread indicates downtrend
- Score captures both direction and magnitude of the crossover gap
- Buy signal: short MA crosses above long MA
- Sell signal: short MA crosses below long MA



Strategy 3: Adaptive Volatility Band

Signal type: continuous

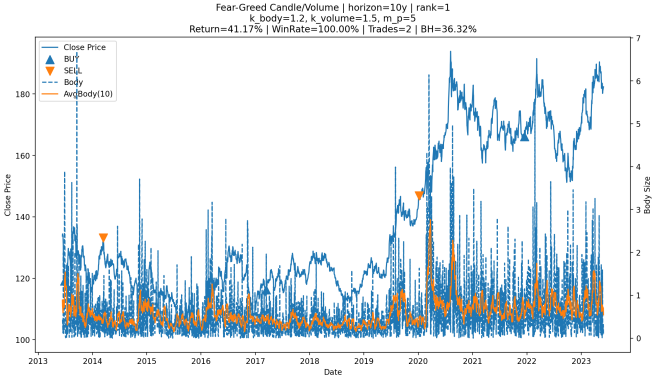
- Uses a volatility proxy derived from the high-low-close range
- Bands are placed relative to the rolling volatility level
- Score reflects the current volatility regime position
- High-vol breakouts and low-vol compressions generate distinct signals
- Complementary to price-band strategies



Strategy 4: Fear-Greed Candle-Volume

Signal type: event-style (discrete)

- Fires only when both candle body size and volume spike simultaneously above rolling thresholds
- Captures extreme sentiment: large directional move confirmed by high activity
- Rare by design: most days produce no signal
- Unlike the other three strategies, this is not a continuous score
- Adds a qualitatively different basis vector to the design matrix



Objective 1: Return-Based Strategy Combination

- Objective 1 combined realized strategy returns.
- For each anchor date:
  - optimize candidate parameters inside each strategy family
  - select representative candidates
  - build a daily return matrix
  - estimate portfolio weights
- Weight constraints tested:
  - signed weights
  - long-only weights

Objective 1 built the rolling optimization infrastructure.

Why Objective 2 Was Needed

- Objective 1 asks:

  How should realized strategy returns be combined?

- Objective 2 asks a different question:

  Do current strategy states predict future returns?

- This moves the project from portfolio mixing to supervised learning.
- The strategy outputs become features.
- Future cumulative return becomes the target.

Objective 2: Initial Design Matrix

$$\hat{y}_t = \beta_0 + A_t \beta$$

- Initial A matrix had four columns:
  - adaptive band score
  - MA crossover score
  - adaptive volatility band score
  - fear-greed score
- Target:

$$y_t^{(h)} = \frac{P_{t+h}}{P_t} - 1$$

- h is the future cumulative-return horizon.

Normal Equation Baseline

- The first benchmark was ordinary least squares:

$$\min_{\beta} \|y - A\beta\|^2$$

- The classical normal-equation form is:

$$\hat{\beta} = (A^\top A)^{-1} A^\top y$$

- In code, this is solved with stable numerical least squares.
- This made the strategy-space idea directly comparable to standard linear supervised learning.

Horizon Scan

- Instead of predicting only the next day, we scanned:

$$h = 1, 2, \ldots, 130$$

- The target is not the return on day h.
- It is the full cumulative return from today to h trading days later:

$$y_t^{(h)} = \frac{P_{t+h}}{P_t} - 1$$

- This asks which future time scale the strategy space naturally explains.

Expanded Strategy Space

- The initial 4-column space was expanded to 40 basis columns.
- Construction:
  - top 10 adaptive band candidates
  - top 10 MA crossover candidates
  - top 10 adaptive volatility candidates
  - top 10 fear-greed candidates
- This is closer to the class model-selection exercise:

$$\text{many basis functions} \rightarrow \text{choose the useful structure}$$

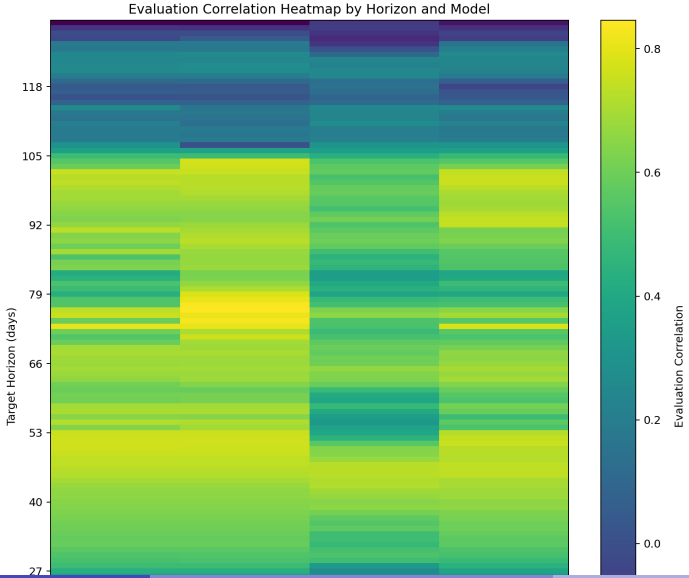
- Models compared:

$$\text{OLS, Ridge, Lasso, Elastic Net}$$

Forward Validation Design

- For each anchor date, only the anchor's known optimization snapshot is used.
- Build the 40-column strategy matrix from that snapshot.
- Fit models over the selection period.
- Evaluate on the following future period.
- This separates:
  - model discovery
  - forward validation
  - investment translation

Horizon-Model Correlation Map



Best Horizon by Anchor

Anchor	Best Horizon	Best Model	Eval Corr
2020-06-30	120	OLS	0.4054
2020-12-31	61	OLS	0.8371
2021-06-30	73	OLS	0.6700
2021-12-31	110	OLS	0.7314
2022-06-30	127	OLS	0.9065
2022-12-31	126	OLS	0.6676
2023-06-30	128	OLS	0.8597
2023-12-31	122	OLS	0.6098
2024-06-30	127	OLS	0.7046
2024-12-31	45	OLS	0.7320

Main finding: the predictive horizon is regime-dependent.

Interpretation of Horizon Shifts

- Long horizons near 120–130 days appeared when GLD moved with slower, persistent drift.
- Medium horizons near 45–73 days appeared when transitions or acceleration mattered more.
- This suggests that the model is not only learning a strategy signal.
- It is also learning the natural time scale of the current market regime.

Practical interpretation:

$$\text{slow trend regime} \rightarrow \text{long horizon}$$

$$\text{transition regime} \rightarrow \text{shorter horizon}$$

From Prediction to Investment

- The model output is a predicted future cumulative return.
- It is converted into a long-only target weight:

$$w_t = \text{clip}\left(\frac{\hat{y}_t}{s}, 0, 1\right)$$

- Then the portfolio is split into rolling tranches.
- Each day:
  - compute the signal
  - allocate one tranche based on w_t
  - hold the tranche for the fixed horizon

Tranche Interpretation

- A 45-day tranche is more responsive.
- A 130-day tranche is slower and more defensive.
- This translates the predictive horizon into an executable allocation rule.

Tranche	Behavior	Practical Role
45-day	faster reaction	transition capture
130-day	slower reaction	defensive long-horizon filter

Update Frequency Experiment

- The next question was operational:

How often should the model be refreshed?

- We compared:

$$1M,\ 2M,\ 3M,\ 6M$$

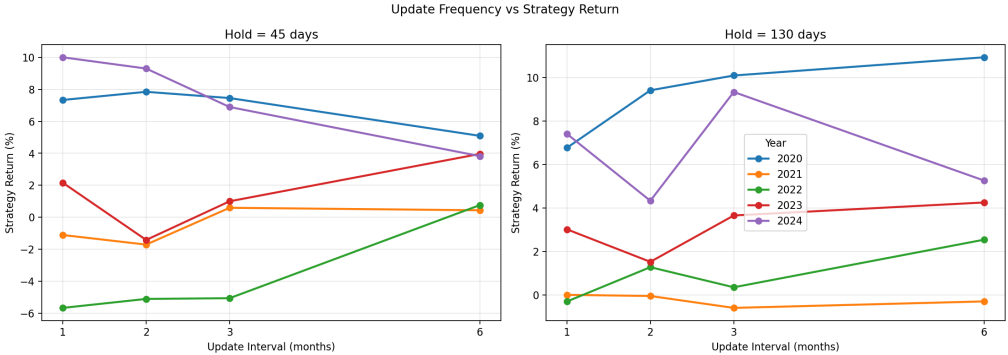
- This tests the trade-off between:
  - fast adaptation
  - noise chasing
  - regime stability

Update Frequency Summary

Year	Buy and Hold	45d Best	130d Best
2020	20.22%	2M best	6M raw, 3M efficient
2021	-6.24%	3M best	1M best
2022	0.78%	6M best	6M best
2023	11.76%	6M best	6M best
2024	26.96%	1M best	3M best

- No single update frequency dominates all years.
- 1M behaves like a fast adaptive mode.
- 3M behaves like a balanced compromise.
- 6M works best in persistent regimes.

Update Frequency: Return by Year



Each line is one calendar year. No single update frequency dominates across all years.

Capital Efficiency View

- Absolute return is not the only metric.
- In a multi-asset system, unused capital can be allocated elsewhere.
- So we also track exposure-adjusted efficiency:

$$100\%\text{-Eq Return} \approx \frac{\text{Strategy Return}}{\text{Average Exposure}}$$

- Across 40 tested runs:
  - raw return beat buy-and-hold in 10 cases
  - efficiency proxy beat buy-and-hold in 17 cases

What Still Needs Work

- The model is not yet a stable regime-aware exposure controller.
- Block analysis showed mixed results:
  - sometimes exposure rose in good blocks
  - sometimes it also rose in weak blocks
- This suggests the next frontier is not only predicting return.
- It is learning when exposure itself should be high or low.

Live Deployment Direction

- Deployment target: GLD through Alpaca paper trading.
- Daily flow:
  1. update GLD daily data
  2. compute latest strategy-space signal
  3. build target position
  4. submit or log the order
  5. send email alert
- Raspberry Pi runs the daily pipeline with a systemd timer.

Final Takeaways

- Strategy outputs can be treated as evaluated basis functions.
- This creates a valid numerical strategy space.
- Model selection on this space reveals horizon and regime structure.
- Tranche simulation translates prediction into allocation.
- The long-term goal is a multi-asset allocation system, not only a GLD strategy.

Strategy-space modeling turns interpretable trading rules into a testable predictive geometry.